Clustering

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Dependencies

This document depends on the following packages:

```
library(devtools)
library(Biobase)
library(dendextend)
```

To install these packages you can use the code (or if you are compiling the document, remove the `eval=FALSE` from the chunk.)

```
install.packages(c("devtools","dendextend"))
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("Biobase"))
```

General principles

- How do we define close?
- How do we group things?
- How do we visualize the grouping?
- How do we interpret the grouping?

Load some data

We will use this expression set to look at how we use plots and tables to check for different characteristics

```
con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bodymap_eset.RData")
load(file=con)
close(con)
bm = bodymap.eset
pdata=pData(bm)
edata=as.data.frame(exprs(bm))
fdata = fData(bm)
ls()
```

```
## [1] "bm"          "bodymap.eset" "con"          "edata"
## [5] "fdata"       "pdata"        "tropical"
```

Calculate distances between samples

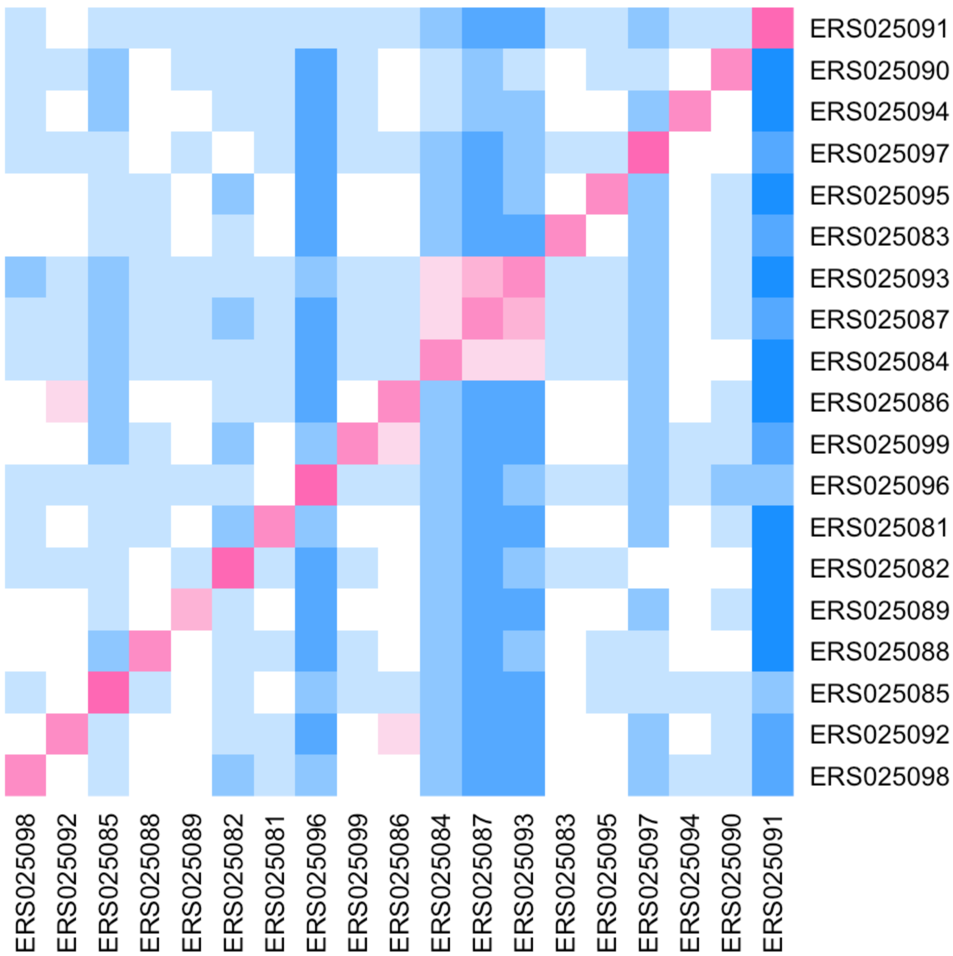
- Most important step
- Garbage in -> garbage out
- Distance or similarity
- Continuous - euclidean distance
- Continous - correlation similarity
- Binary - manhattan distance
- Pick a distance/similarity that makes sense for your problem

First we log transform and remove lowly expressed genes, then calculate Euclidean distance

```
edata = edata[rowMeans(edata) > 5000,]
edata = log2(edata + 1)

# By default calculates the distance between rows
dist1 = dist(t(edata))

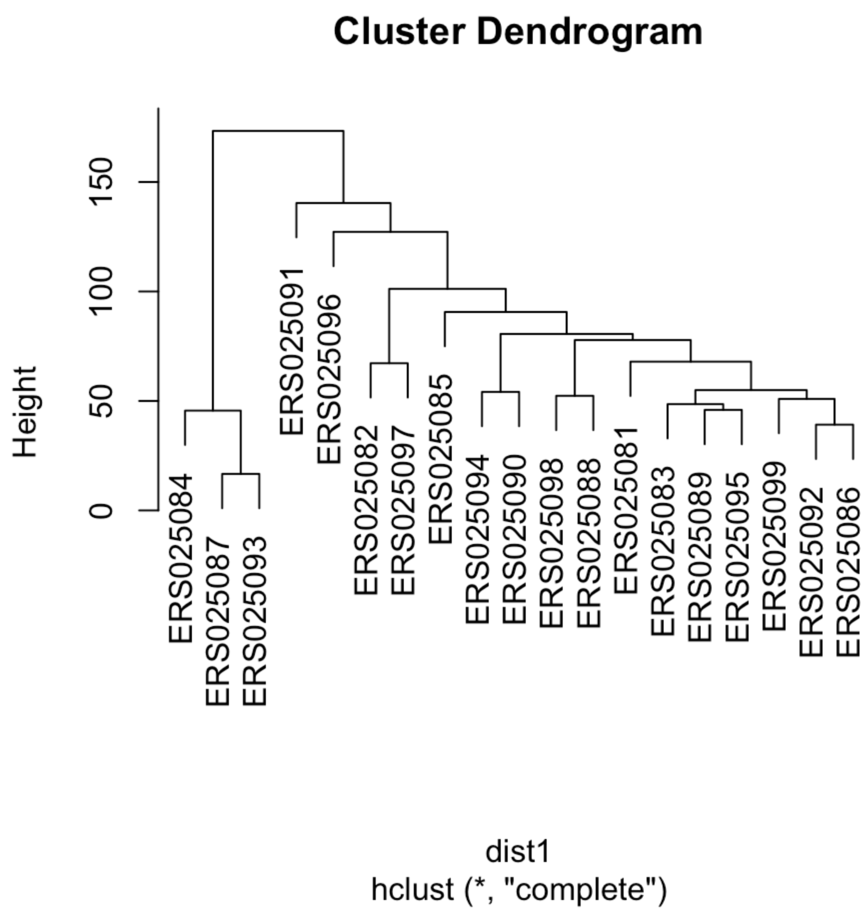
## Look at distance matrix
colramp = colorRampPalette(c(3,"white",2))(9)
heatmap(as.matrix(dist1),col=colramp,Colv=NA,Rowv=NA)
```



Now cluster the samples

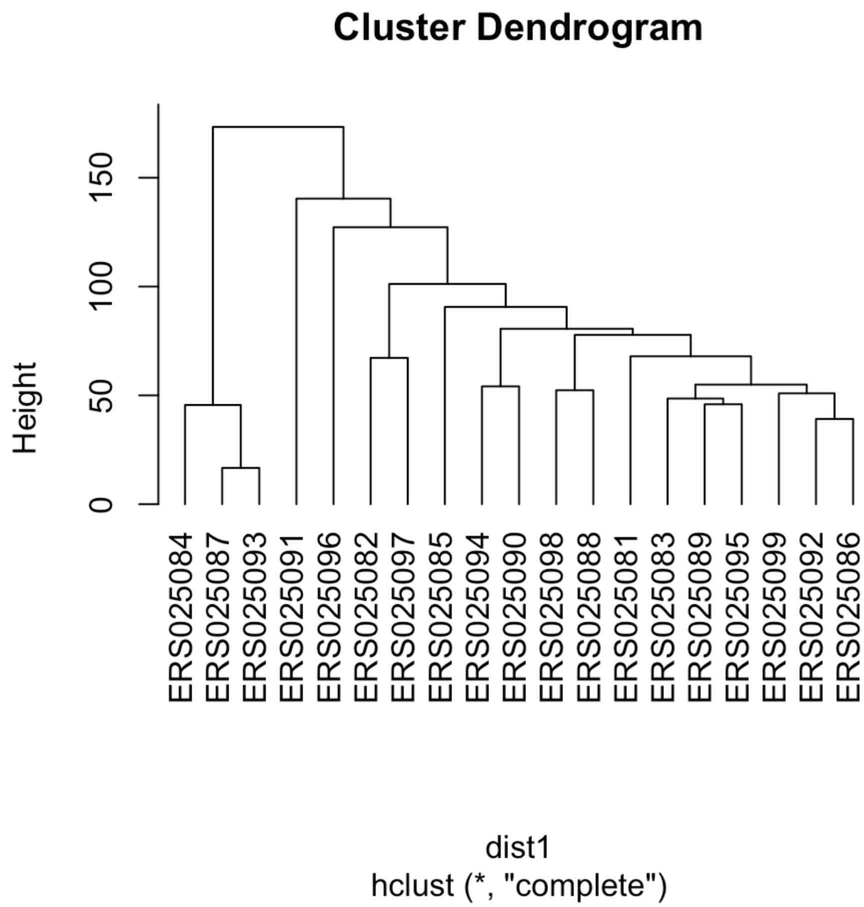
Here we use the distance we previously calculated to perform a hierarchical clustering and plot the dendrogram:

```
hclust1 = hclust(dist1)
plot(hclust1)
```



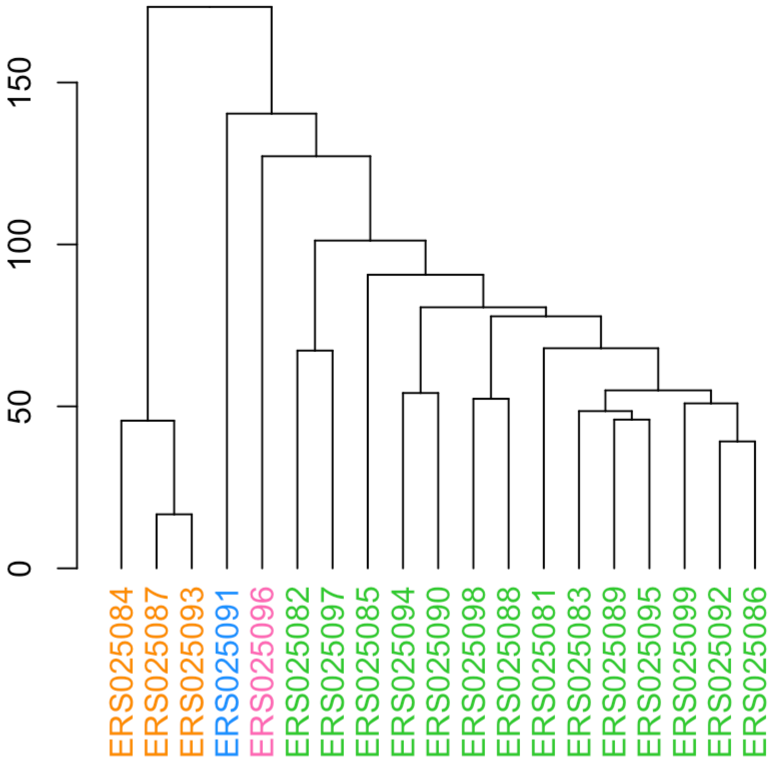
We can also force all of the leaves to terminate at the same spot

```
plot(hclust1,hang=-1)
```



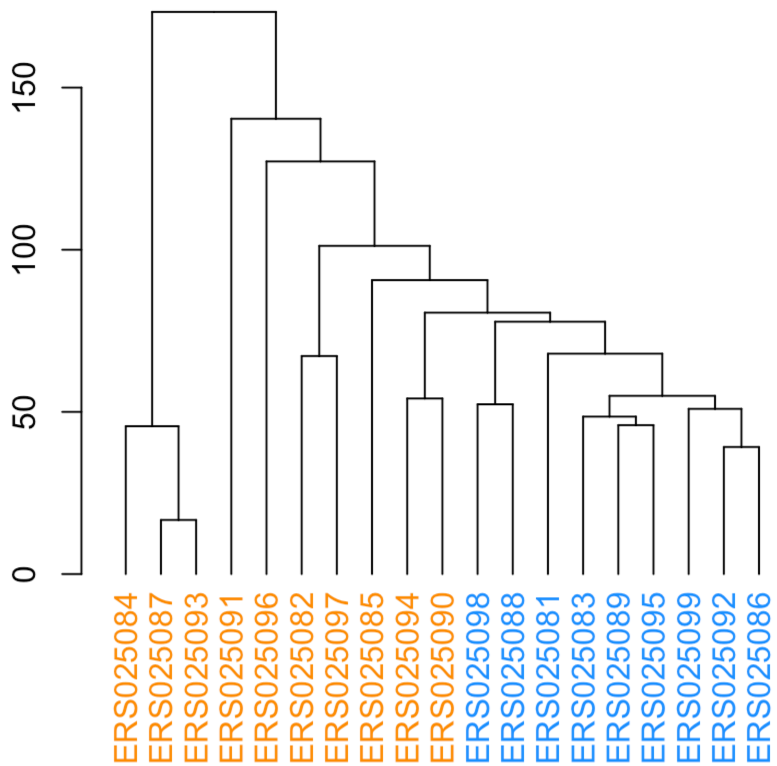
We can also color the dendrogram either into a fixed number of groups

```
dend = as.dendrogram(hclust1)
dend = color_labels(hclust1,4,col=1:4)
plot(dend)
```



Or you can color them directly

```
labels_colors(dend) = c(rep(1,10),rep(2,9))
plot(dend)
```



This can get very fancy and we won't cover all the options here but for further ideas check out the dendextend package (http://htmlpreview.github.io/?https://github.com/talgalili/dendextend/blob/master/inst/ignored/Introduction%20to%20dendextend.html).

K-means clustering

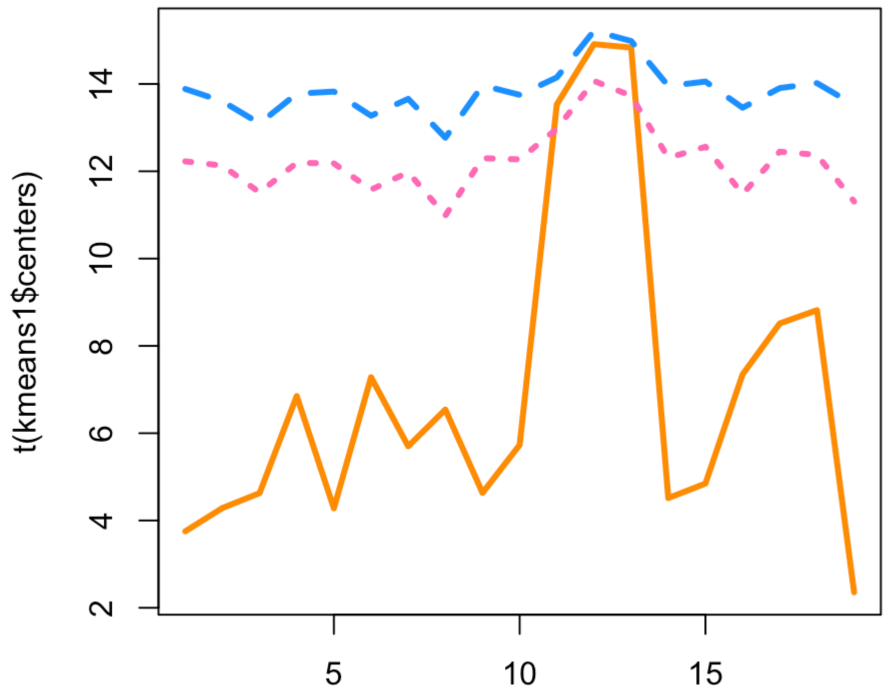
Now we can perform k-means clustering. By default, the rows are clustered. You can either input the cluster means (often unknown) or the number of clusters (here I input 3 clusters)

```
kmeans1 = kmeans(edata,centers=3)
names(kmeans1)
```

```
## [1] "cluster"      "centers"      "totss"        "withinss"
## [5] "tot.withinss" "betweenss"    "size"         "iter"
## [9] "ifault"
```

Now we can look at the cluster centers

```
matplot(t(kmeans1$centers),col=1:3,type="l",lwd=3)
```



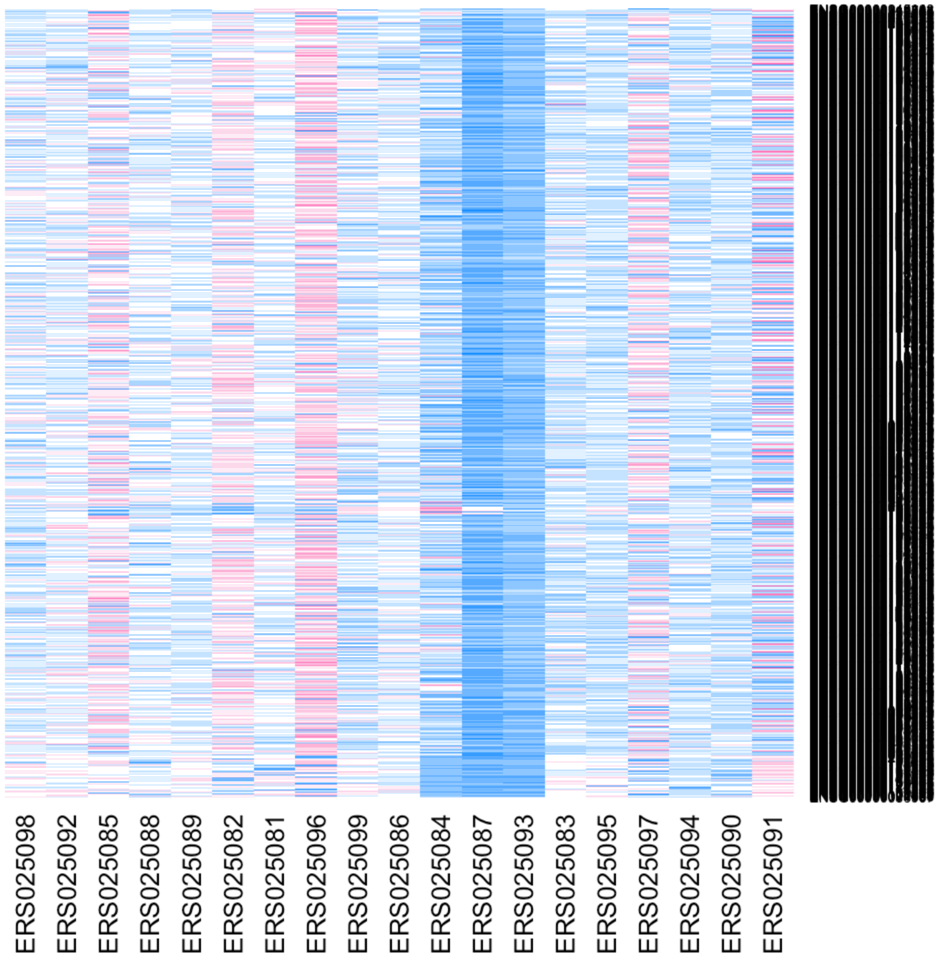
We can also look at how many belong to each cluster

```
table(kmeans1$cluster)
```

```
##
##   1   2   3
##  63 390 745
```

And cluster the data together and plot

```
heatmap(as.matrix(edata)[order(kmeans1$cluster),],col=colramp,Colv=NA,Rowv=NA)
```



Note that this is a random start so you get a different clustering if you run it again you get a different answer

```
kmeans2 = kmeans(edata,centers=3)
table(kmeans1$cluster,kmeans2$cluster)
```

```
##
##       1   2   3
##   1   3  60   0
##   2  15   0 375
##   3 151   0 594
```

Session information

Here is the session information

```
devtools::session_info()
```

```
##  setting  value
##  version  R version 3.2.1 (2015-06-18)
##  system   x86_64, darwin10.8.0
##  ui       RStudio (0.99.447)
##  language (EN)
##  collate  en_US.UTF-8
##  tz       America/New_York
##
##  package           * version     date
##  acepack             1.3-3.3     2014-11-24
##  annotate            1.46.1      2015-07-11
##  AnnotationDbi     * 1.30.1      2015-04-26
##  assertthat          0.1         2013-12-06
##  BiasedUrn         * 1.06.1      2013-12-29
##  Biobase           * 2.28.0      2015-04-17
##  BiocGenerics      * 0.14.0      2015-04-17
##  BiocInstaller     * 1.18.4      2015-07-22
##  BiocParallel        1.2.20      2015-08-07
##  biomaRt             2.24.0      2015-04-17
##  Biostrings          2.36.3      2015-08-12
##  bitops              1.0-6       2013-08-17
##  bladderbatch      * 1.6.0       2015-08-26
##  broom             * 0.3.7       2015-05-06
##  caTools             1.17.1      2014-09-10
##  cluster             2.0.3       2015-07-21
##  colorspace          1.2-6       2015-03-11
##  corpcor             1.6.8       2015-07-08
##  curl                0.9.2       2015-08-08
##  DBI               * 0.3.1       2014-09-24
##  dendextend        * 1.1.0       2015-07-31
##  DESeq2            * 1.8.1       2015-05-02
##  devtools          * 1.8.0       2015-05-09
##  digest              0.6.8       2014-12-31
##  dplyr             * 0.4.3       2015-09-01
##  edge              * 2.1.0       2015-09-06
##  evaluate            0.7.2       2015-08-13
##  foreign             0.8-66      2015-08-19
##  formatR             1.2         2015-04-21
##  Formula           * 1.2-1       2015-04-07
##  futile.logger       1.4.1       2015-04-20
##  futile.options      1.0.0       2010-04-06
##  gdata               2.17.0      2015-07-04
##  genefilter        * 1.50.0      2015-04-17
##  geneLenDataBase   * 1.4.0       2015-09-06
##  geneplotter         1.46.0      2015-04-17
##  GenomeInfoDb      * 1.4.2       2015-08-15
##  GenomicAlignments   1.4.1       2015-04-24
##  GenomicFeatures     1.20.2      2015-08-14
##  GenomicRanges     * 1.20.5      2015-06-09
##  genstats          * 0.1.02      2015-09-05
##  ggplot2           * 1.0.1       2015-03-17
##  git2r               0.11.0      2015-08-12
```

```
##  GO.db                3.1.2       2015-09-06
##  goseq              * 1.20.0      2015-04-17
##  gplots             * 2.17.0      2015-05-02
##  gridExtra            2.0.0       2015-07-14
##  gtable               0.1.2       2012-12-05
##  gtools               3.5.0       2015-05-29
##  highr                0.5         2015-04-21
##  HistData           * 0.7-5       2014-04-26
##  Hmisc              * 3.16-0      2015-04-30
##  htmltools            0.2.6       2014-09-08
##  httr                 1.0.0       2015-06-25
##  IRanges            * 2.2.7       2015-08-09
##  KernSmooth           2.23-15     2015-06-29
##  knitr              * 1.11        2015-08-14
##  lambda.r             1.1.7       2015-03-20
##  lattice            * 0.20-33     2015-07-14
##  latticeExtra         0.6-26      2013-08-15
##  lazyeval             0.1.10      2015-01-02
##  limma              * 3.24.15     2015-08-06
##  lme4                 1.1-9       2015-08-20
##  locfit               1.5-9.1     2013-04-20
##  magrittr             1.5         2014-11-22
##  MASS               * 7.3-43      2015-07-16
##  Matrix             * 1.2-2       2015-07-08
##  MatrixEQTL         * 2.1.1       2015-02-03
##  memoise              0.2.1       2014-04-22
##  mgcv               * 1.8-7       2015-07-23
##  minqa                1.2.4       2014-10-09
##  mnormt               1.5-3       2015-05-25
##  munsell              0.4.2       2013-07-11
##  nlme               * 3.1-122     2015-08-19
##  nloptr               1.0.4       2014-08-04
##  nnet                 7.3-10      2015-06-29
##  org.Hs.eg.db       * 3.1.2       2015-07-17
##  plyr                 1.8.3       2015-06-12
##  preprocessCore     * 1.30.0      2015-04-17
##  proto                0.3-10      2012-12-22
##  psych                1.5.6       2015-07-08
##  qvalue             * 2.0.0       2015-04-17
##  R6                   2.1.1       2015-08-19
##  RColorBrewer         1.1-2       2014-12-07
##  Rcpp               * 0.12.0      2015-07-25
##  RcppArmadillo      * 0.5.400.2.0 2015-08-17
##  RCurl                1.95-4.7    2015-06-30
##  reshape2             1.4.1       2014-12-06
##  rmarkdown            0.7         2015-06-13
##  rpart                4.1-10      2015-06-29
##  Rsamtools            1.20.4      2015-06-01
##  RSkittleBrewer     * 1.1         2015-09-05
##  RSQLite            * 1.0.0       2014-10-25
##  rstudioapi           0.3.1       2015-04-07
##  rtracklayer          1.28.9      2015-08-19
```

```
##  rversions           1.0.2      2015-07-13
##  S4Vectors         * 0.6.5      2015-09-01
##  scales              0.3.0      2015-08-25
##  snm                 1.16.0     2015-04-17
##  snpStats          * 1.18.0     2015-04-17
##  stringi             0.5-5      2015-06-29
##  stringr             1.0.0      2015-04-30
##  survival          * 2.38-3     2015-07-02
##  sva               * 3.14.0     2015-04-17
##  tidyr               0.2.0      2014-12-05
##  UsingR            * 2.0-5      2015-08-06
##  whisker             0.3-2      2013-04-28
##  XML                 3.98-1.3   2015-06-30
##  xml2                0.1.2      2015-09-01
##  xtable              1.7-4      2014-09-12
##  XVector             0.8.0      2015-04-17
##  yaml                2.1.13     2014-06-12
##  zlibbioc            1.14.0     2015-04-17
##  source
##  CRAN (R 3.2.0)
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##  CRAN (R 3.2.2)
##  Github (jdstorey/edge@a1947b5)
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##  CRAN (R 3.2.1)
##  CRAN (R 3.2.1)
##  Bioconductor
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##  CRAN (R 3.2.0)
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It is also useful to compile the time the document was processed. This document was processed on: 2015-09-06.